Introduction

The parallel-group randomised controlled trial (RCT) is the gold standard for evaluating interventions. Participants are randomly allocated to one of two (or more) arms – typically a new intervention vs control – and followed for a pre-specified outcome. Randomisation balances known and unknown confounders in expectation; blinding further reduces bias.

Prerequisites

Randomisation; hypothesis testing; sample-size calculation.

Theory

Essential elements: - **Primary outcome** (continuous, binary, time-to-event) with a clinically meaningful effect size. - **Allocation ratio** (usually 1:1, occasionally 2:1 or 3:1 for rare interventions). - **Randomisation list** (pre-generated, concealed at allocation time). - **Blinding** (single, double, triple). - **Pre-registered statistical analysis plan**.

Primary analysis is typically intent-to-treat (ITT), comparing groups as randomised.

Assumptions

Participants are exchangeable post-randomisation; allocation is fully concealed; outcome assessment is blinded; follow-up is complete or MAR.

R Implementation

```r
library(pwr)

# Sample-size for two-arm comparison of means
# Effect: difference of 0.5 SD, alpha = 0.05, power = 0.80
pwr.t.test(d = 0.5, sig.level = 0.05, power = 0.80,
           type = "two.sample", alternative = "two.sided")

# Simulate an RCT with continuous outcome
set.seed(2026)
n_per_arm <- 64
arm <- rep(c("ctrl", "trt"), each = n_per_arm)
y   <- rnorm(2 * n_per_arm, mean = ifelse(arm == "trt", 0.5, 0))

# ITT analysis: two-sample t-test
t.test(y ~ arm, var.equal = TRUE)

# Adjusted analysis with a pre-specified covariate (ANCOVA)
covar <- rnorm(2 * n_per_arm)
summary(lm(y ~ arm + covar))$coefficients
```

Output & Results

Sample size ~64 per arm for 80 % power; ITT t-test recovers the effect; ANCOVA-adjusted analysis gives a similar point estimate with smaller SE when the covariate is prognostic.

Interpretation

"The trial randomised 128 participants 1:1 to intervention vs control; the intervention arm showed a 0.52 SD improvement (95 % CI 0.18-0.86, $p = 0.003$) on the primary outcome, analysed by ANCOVA adjusting for baseline value."

Practical Tips

- Register the protocol and SAP **before** enrolment (clinicaltrials.gov, EUDRACT).
- Pre-specify the primary outcome and analysis; secondary outcomes are exploratory.
- Blinding is protective; document how it was broken (unblinding events, assessment).
- Report per CONSORT 2010 guidelines; flow diagram is mandatory.
- ITT is the primary analysis; supportive per-protocol analyses are sensitivity.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale